

Reward-Centered Continuing Sarsa

Status: independent main proposal with completed extended reward-shift/alpha sweep.

Abstract

This proposal studies a basic invariance requirement for continual reinforcement learning: in a continuing control task, adding a constant to every reward should not change the task-relevant behavior. Ordinary discounted action-value learning does not automatically respect this invariance because reward offsets appear in the scale of TD errors and action values. We test whether online reward centering and differential Sarsa reduce this arbitrary reward-origin sensitivity in a continuing access-control queue. The extended run shows that discounted Sarsa's action-value norm grows sharply across reward shifts and alphas, while reward-centered and differential variants keep value scale much more stable and preserve unshifted reward. The study is not framed as "one algorithm scores higher"; it asks whether a continuing agent can learn in units that do not depend on an arbitrary reward baseline.

Standalone Study Summary

This study tests reward-shift invariance in continuing control. The RL problem is the access-control queue: the agent observes server availability and customer priority, then accepts or rejects each customer while learning online. The implemented methods are discounted Sarsa, reward-centered Sarsa, and differential Sarsa with linear/tabular action values. The experiment varies constant reward shifts and measures unshifted reward, accept behavior, high-priority acceptance, reward baseline, TD error, and Q norm. The main evidence is now a 20-seed, 20000-step alpha sweep: ordinary discounted Sarsa develops large reward-shift-dependent value scales, while centered and differential variants keep value norms and unshifted reward more stable. The next step is beta/gamma sensitivity and a midstream reward-origin switch.

Proposal Template Answers

Focused RL question: In an online continuing control task, can an agent learn action values whose scale and behavior are insensitive to an arbitrary constant added to every reward? The question is not whether reward-centered Sarsa has the highest shifted return, but whether the learning dynamics respect the reward-origin invariance expected in an average-reward continuing problem.

Setting and testbed: The testbed is the access-control queue from tabular continuing RL. It is larger and more meaningful than a two-state diagnostic because it has state-dependent action feasibility, priority-dependent rewards, and a real accept/reject control tradeoff, while still being interpretable enough to audit TD errors and value norms.

Implemented comparison: The implemented comparison is discounted Sarsa versus reward-centered Sarsa versus differential Sarsa, all trained online from a single stream with no replay buffer and no deep network. The varied factors are reward shift and alpha in the completed run; beta, gamma, and midstream reward-origin changes are the required next factors.

Observation or figure that answers the question: The main figures must jointly show unshifted reward, Q norm, high-priority acceptance, and divergence as a function of reward shift. The proposal is supported only if centered methods maintain behavior while preventing value-scale inflation; a reward-only table would not answer the

invariance question.

Compute need and fallback: The current extended evidence is CPU-scale and already complete for the alpha/reward-shift grid. If there is no time for a larger sweep, the honest fallback is to submit this as a strong fixed-condition invariance study and state that nonstationary reward-origin switching remains future work.

Independent Research Scope

This proposal is an independent study of reward-origin invariance in continuing control. It should be read separately from Output-Controlled TD and Scale-Invariant Continuing Control: those reports study feature-scale and combined-unit effects, while this report isolates the reward side of the problem. The study deliberately does not claim to solve all continuing average-reward learning, all reward shaping, or all nonstationarity. Its scope is narrower and sharper: a constant reward translation should not force a retuned step size, distorted value scale, or changed accept/reject behavior in access-control Sarsa.

The proposal also has a clear boundary relative to the small Centered TD Diagnostics report. The diagnostic report explains the mechanism in a tiny prediction setting; this report supplies the control environment, policy probes, seed sweep, and main evidence. If only one reward-centering proposal is submitted, this one should be the main paper-style study and the diagnostic should be cited as supporting material.

Evidence Level

Evidence level: strong independent main-candidate evidence, with specific unfinished sensitivity tests. The completed result uses 20 seeds, 20000 online steps per condition, five reward shifts, three alphas, and three algorithms. This is substantially stronger than a pilot and is enough to support the central fixed-condition claim that ordinary discounted Sarsa is reward-origin sensitive while centered/differential variants are much less sensitive.

The evidence is not yet a full continual-adaptation paper. It does not include a no-reset midstream reward-origin switch, a beta sweep for the reward baseline, or a gamma sweep for discounted variants. Those omissions do not invalidate the current fixed-condition result, but they limit the conclusion to reward-origin invariance across separate streams rather than adaptation to a changing reward sensor inside one stream.

Research Motivation

The Alberta Plan frames intelligence as a temporally uniform stream of experience: the agent is always acting, always learning, and not periodically reset into a special training phase. In that setting, reward origin is an especially clean stress test. If every reward is translated by a constant, the preference ordering of policies in an average-reward continuing task should not change. A long-lived agent should therefore not require a new step size or a new representation merely because an engineer redefined the zero point of the reward sensor.

Discounted value methods can violate this practical invariance. With gamma near one, a constant reward shift contributes a large constant component to the value function. That component is not useful for choosing actions, but it changes TD-error magnitudes, value norms, and sometimes behavior through function approximation and finite step-size effects. Reward centering is attractive because it is small, online, and compatible with the course constraints: no replay buffer, no deep network, and no offline fitting.

Research Question

Does online reward centering make Sarsa robust to constant reward shifts in a non-episodic continuing control task?

The precise hypothesis is:

> Reward-centered Sarsa and differential Sarsa should show lower variation in value scale and unshifted task reward across reward shifts than ordinary discounted Sarsa.

This hypothesis is about invariance and learning dynamics, not about final shifted return. All primary results therefore report unshifted reward separately from the shifted reward observed by the learner.

Alberta Plan Connection

This proposal targets several Alberta Plan themes:

- - continuing problems rather than episodic reset tasks;
- - value functions as central predictive/control quantities;
- - average-reward thinking for long-lived agents;
- - online adaptation from ordinary experience;
- - small mechanisms that reduce sensitivity to hand-tuned scales.

The task is deliberately not deep RL. The point is to inspect a Core RL mechanism with enough transparency that value-scale failures can be explained.

Related Work

The main source is Reward Centering, which shows that subtracting an empirical average reward improves continuing discounted methods and removes constant reward-shift sensitivity. Average-reward RL and the access-control queue provide the canonical setting. Bellman Error Centering gives a more recent theoretical lens on related centered fixed points. The Alberta Plan supplies the broader motivation for continuing agents and temporal uniformity.

Local references:

- - [resources/alberta_plan_related/reward_centering_2405.09999.pdf](#)
- - [resources/alberta_plan_related/bellman_error_centering_2502.03104.pdf](#)
- - [resources/alberta_plan_related/average_reward_learning_planning_2006.16318.pdf](#)
- - [resources/alberta_plan_related/alberta_plan_2208.11173.pdf](#)

Environment

The main environment is an access-control queue. The agent observes:

- - number of free servers;
- - priority class of the current customer.

The agent chooses:

- - reject;
- - accept if a server is free.

The unshifted reward is the accepted customer's priority, or zero for rejection. The experimental observed reward is:

$\text{observed_reward} = \text{unshifted_reward} + \text{reward_shift}.$

This creates several equivalent descriptions of the same decision problem. The agent must learn online from a single stream with no replay.

Methods

The study compares three on-policy control agents:

- - Discounted Sarsa with gamma 0.99.
- - Reward-centered discounted Sarsa with an online average-reward estimate.
- - Differential Sarsa with gamma 1.0 and average-reward correction.

All methods use tabular one-hot state features, epsilon-greedy behavior, and one update per environment step. Reward centering updates an average reward estimate online and subtracts that estimate from the immediate reward term. Differential Sarsa uses the average-reward style TD error.

The update details are deliberately kept explicit because the proposal is about the unit of the learning signal. Ordinary discounted Sarsa uses $\delta = R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$. Reward-centered Sarsa uses $\delta = (R_{t+1} - r_{\text{bar}_t}) + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$ with an online baseline update $r_{\text{bar}} \leftarrow r_{\text{bar}} + \beta (R_{t+1} - r_{\text{bar}})$. Differential Sarsa uses the average-reward form $\delta = R_{t+1} - r_{\text{bar}_t} + Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)$ and updates the reward baseline from the same continuing stream. The report logs observed shifted reward and unshifted task reward separately, so the evaluation is not accidentally rewarding the algorithm for seeing a larger arbitrary constant.

Experimental Design

Current extended sweep:

- - Environment: access-control queue.
- - Reward shifts: -8, -4, 0, 4, 8.
- - Alphas: 0.02, 0.05, 0.1.
- - Seeds: 0-19.
- - Steps: 20000 per condition.
- - Algorithms: discounted Sarsa, reward-centered Sarsa, differential Sarsa.
- - Result path: `experiments/alberta_core_rl/results/reward_centered_sarsa/20260709T024517Z_extended`.

Primary metrics:

- - unshifted average reward;
- - Q norm;
- - TD-error scale;
- - reward-bar estimate;
- - accept rate;
- - simple policy probes.

The decision rule is: a method supports the hypothesis only if it preserves unshifted reward and policy probes while keeping Q norm and TD-error scale approximately invariant across reward shifts. A method that earns high shifted reward but changes behavior, explodes in value norm, or needs a different alpha for each shift does not count as solving the continuing reward-origin problem.

Remaining sensitivity tests:

- - Beta values: 0.003, 0.01, 0.03.

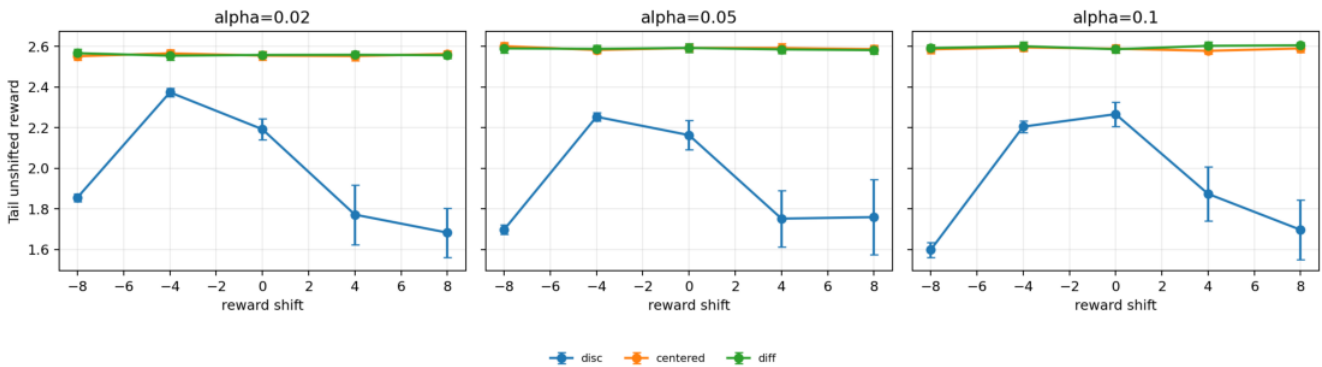
- - Gamma values for discounted and centered variants.
- - Nonstationary extension: switch reward origin midway through a single stream without resetting weights, then measure reward-bar lag and recovery window.

The report-ready figures below use seed-tail condition summaries rather than dense learning-curve overlays. They show reward, value scale, behavior, and divergence separately, which is the right evidence for an invariance claim.

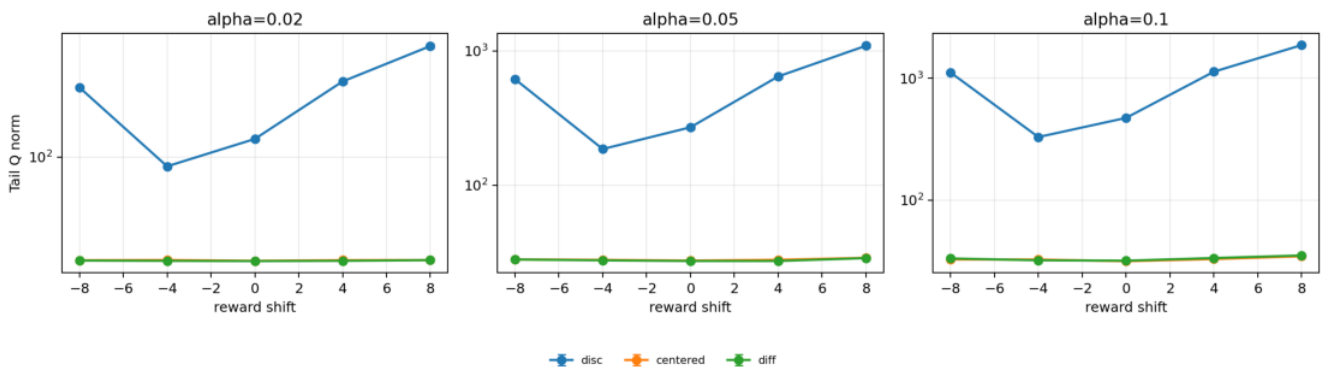
Experiment Design Rationale

The environment, variables, and metrics were chosen to separate three explanations that a skeptical reviewer would otherwise conflate. First, unshifted reward and high-priority acceptance test task behavior rather than the shifted scalar reward observed by the learner. Second, Q norm and TD-error scale test whether the learner has encoded an arbitrary reward offset as a large value component. Third, divergence and policy probes test whether the effect is merely cosmetic or can change actual control. The access-control queue is therefore not being used as a benchmark leaderboard; it is being used as a controlled continuing system in which reward-origin invariance can be measured.

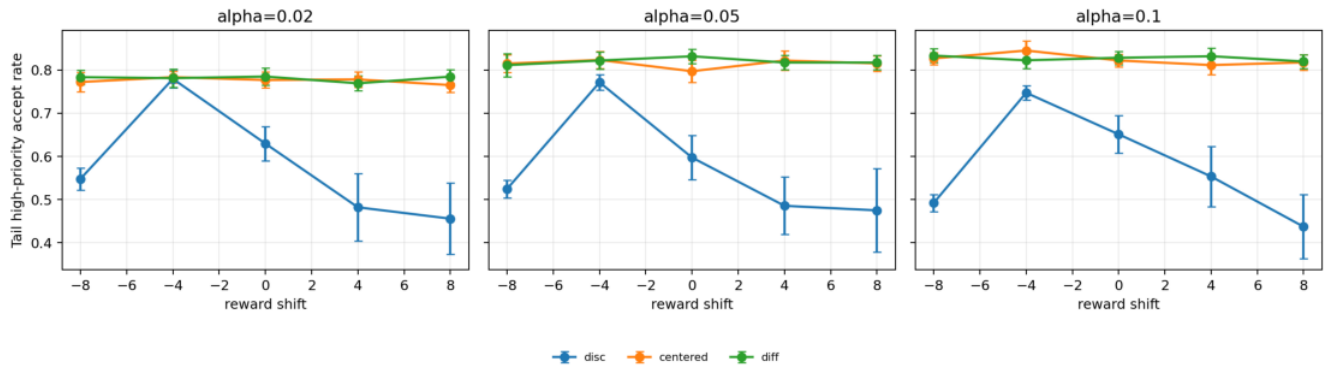
The current grid is deliberately modest rather than blind. Reward shifts -8 through 8 are large relative to access-control priorities, so they stress the nuisance value offset. The alpha grid includes settings where ordinary Sarsa still learns and settings where value-scale inflation becomes severe. The next experiment should not simply add more random hyperparameters; it should target the remaining causal gaps: beta controls baseline lag, gamma controls discounted offset amplification, and a midstream shift tests continual recovery without resetting weights.



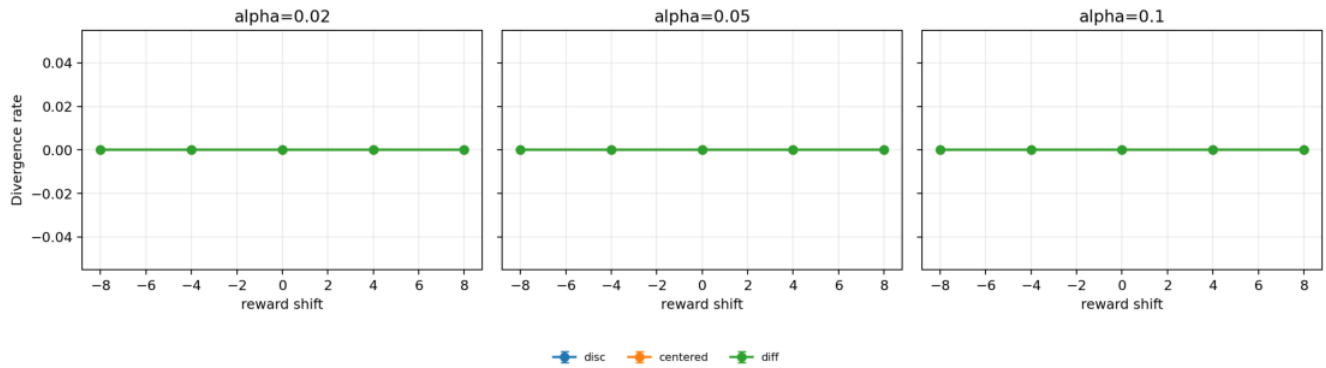
Tail unshifted reward under reward shifts and alpha values.



Tail action-value norm under reward shifts and alpha values.



Tail high-priority accept rate under reward shifts and alpha values.



Divergence rate under reward shifts and alpha values.

Results

The main result is value-scale invariance under a larger seed and alpha sweep. Discounted Sarsa's Q norm grows strongly with reward shift and alpha: at shift 8, alpha 0.1, its tail Q norm is about 1875.55 +/- 38.4, and its unshifted reward is about 1.697 +/- 0.147. This is not useful task knowledge; it is largely the arbitrary value offset induced by the shifted reward stream.

Reward-centered Sarsa keeps Q norm much smaller, roughly 22-35 across the tested reward shifts and alphas, with zero divergence. Its unshifted reward remains around 2.55-2.60 across reward shifts. Differential Sarsa shows the same qualitative scale-stability pattern, with unshifted reward around 2.55-2.61 and similarly modest Q norms. Ordinary discounted Sarsa degrades under large positive and negative shifts, especially at higher alpha.

The high-priority accept-rate figure is included to guard against a misleading reward-only interpretation. Reward centering is not counted as successful merely because it improves a scalar average; it must also preserve the task-relevant accept/reject tendency under reward translations. The divergence panel is mostly a sanity check for this run, but it makes explicit that the key difference is value-scale inflation rather than many hard numerical crashes.

The key interpretation is that reward centering removes most of the nuisance component of the TD target. It does not change the environment or provide extra samples; it changes the learning signal so that the learner focuses on reward deviations rather than reward origin.

Analysis

The result should be read as an invariance test. If a method is robust, changing reward origin should not greatly affect value scale or unshifted behavior. Discounted Sarsa fails this test in the extended sweep. Reward-centered and

differential variants pass it much more closely.

The differential baseline is important. If differential Sarsa performs similarly to reward centering, the conclusion is not that reward centering uniquely solves the problem. The stronger and more honest conclusion is that average-reward-style removal of reward offsets is a necessary design idea for continuing agents. Reward centering is one practical route that preserves a discounted method structure.

Threats To Validity

The current extended result uses 20 seeds and 20000 steps, but it still sweeps only alpha. A high-confidence course-paper version should also sweep beta and gamma because reward-baseline adaptation can be too slow or too noisy.

The environment is canonical but still small and tabular. That is appropriate for Core RL mechanism analysis, but claims should not be generalized to deep agents or large robotics tasks.

The beta step size for the average reward estimator is not fully swept. A slow estimator may lag under nonstationary rewards; a fast estimator may inject variance.

Reward shifts are constant within each run. A stronger continual study should include midstream reward-origin changes without resetting the agent.

Reviewer Critique And Revisions

Sutton-style critique:

- - The report must not claim episodic return improvements; it must focus on continuing invariance and average-reward reasoning.

Experimental critique:

- - The first five-seed pilot was not enough for final statistical claims; this has been addressed with a 20-seed extended sweep.

Baseline critique:

- - Differential Sarsa is a serious competitor, not an appendix baseline.

Revision already made:

- - The initial two-loop MDP was replaced by access-control for the main claim.
- - Shifted observed reward and unshifted task reward are logged separately.

Next required revision:

- - Add beta/gamma ablations and midstream reward-shift changes.

Per-proposal audit matrix:

Reviewer angle	Critique	Action taken	Remaining risk
Alberta Plan	The study must be about continual ordinary experience, not episodic score. Uses a continuing access-control stream and reports unshifted task reward separately from observed reward. Midstream reward-origin switching is still missing.		
Average-reward RL	Discounted Sarsa is not the only relevant baseline. Adds differential Sarsa as a serious average-reward baseline. Gamma/beta interactions need a wider sweep.		
Statistics	Five seeds were insufficient. Completed a 20-seed extended sweep. Confidence intervals are tail summaries; AUC and paired seed effects would strengthen the report.		
Mechanism	Reward improvements alone could hide behavior changes. Adds Q norm, high-priority acceptance, accept rate, and divergence figures. Full policy-distance probes over all states are not yet reported.		
Strict instructor	The proposal should not overclaim a universal reward-centering solution.		

Conclusion is limited to reward-origin invariance in access-control Sarsa. | Larger continuing environments would be needed for a broader claim. |

Conclusion

Reward centering is a strong independent proposal because it asks a clean continuing-RL question and now has extended 20-seed evidence. The result supports the claim that ordinary discounted Sarsa is reward-origin sensitive, while reward-centered and differential variants are far more stable across reward shifts and alphas. The next stage should add beta/gamma sensitivity and nonstationary reward-origin changes so the proposal tests continual adaptation, not only fixed-condition invariance.

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/reward_centered_sarsa/config_main.json
```

Extended seed/step sweep:

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
--config experiments/alberta_core_rl/configs/reward_centered_sarsa/config_extended.json
```

Regenerate the two report figures:

```
cd /mnt/shared-storage-user/yupeng/Core-RL
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/plot_results.py \
--result-dir experiments/alberta_core_rl/results/reward_centered_sarsa/20260709T024517Z_extended \
--y-key avg_unshifted_reward \
--group-keys algorithm reward_shift alpha
```

```
PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
/data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/plot_results.py \
--result-dir experiments/alberta_core_rl/results/reward_centered_sarsa/20260709T024517Z_extended \
--y-key q_norm \
--group-keys algorithm reward_shift alpha
```